Chapter 6 — Moments

Worked Problems and Solutions

Contents

1 Textbook Exercises

Q1 — Book Ch.6, Exercise 13

A fair die is rolled twice, with outcomes X for the first roll and Y for the second roll. Find the moment generating function $M_{X+Y}(t)$ of $X+Y$ (your answer should be a function of t and can contain unsimplified finite sums).

Solution. Since X and Y are i.i.d., the MGF of a sum of independent random variables is the product of the individual MGFs:

$$M_{X+Y}(t) = E\left(e^{t(X+Y)}\right) = E\left(e^{tX}e^{tY}\right) = E\left(e^{tX}\right) E\left(e^{tY}\right).$$

Since X is discrete uniform on $\{1, 2, 3, 4, 5, 6\}$,

$$E\left(e^{tX}\right) = \sum_{k=1}^{6} P(X = k)\, e^{kt} = \frac{1}{6}\sum_{k=1}^{6} e^{kt}.$$

By the identical distribution of Y, $E(e^{tY}) = \frac{1}{6}\sum_{k=1}^{6} e^{kt}$ as well. Therefore

$$M_{X+Y}(t) = \left(\frac{1}{6}\sum_{k=1}^{6} e^{kt}\right)\left(\frac{1}{6}\sum_{k=1}^{6} e^{kt}\right) = \left(\frac{1}{6}\sum_{k=1}^{6} e^{kt}\right)^2.$$

Q2 — Book Ch.6, Exercise 14

Let $U_1, U_2, \ldots, U_{60}$ be i.i.d. Unif$(0, 1)$ and $X = U_1 + U_2 + \cdots + U_{60}$. Find the MGF of X.

Solution. First find the MGF of a single $U_1 \sim$ Unif$(0, 1)$. For $t \neq 0$:

$$E\left(e^{tU_1}\right) = \int_0^1 e^{tu}\, du = \left[\frac{e^{tu}}{t}\right]_0^1 = \frac{e^t}{t} - \frac{e^0}{t} = \frac{e^t - 1}{t}.$$

For $t = 0$: $E(e^{0 \cdot U_1}) = E(1) = 1$.

Since $U_1, \ldots, U_{60}$ are independent, the MGF of the sum is the product of the individual MGFs:

$$E\left(e^{tX}\right) = E\left(e^{t(U_1+\cdots+U_{60})}\right) = \prod_{i=1}^{60} E\left(e^{tU_i}\right) = \left(E\left(e^{tU_1}\right)\right)^{60}.$$

Substituting the single-variable MGF, for $t \neq 0$:

$$E\left(e^{tX}\right) = \left(\frac{e^t - 1}{t}\right)^{60} = \frac{(e^t - 1)^{60}}{t^{60}},$$

and the value is 1 for $t = 0$.

Q3 — Book Ch.6, Exercise 20

Let $X \sim \text{Pois}(\lambda)$, and let $M(t)$ be the MGF of X. The *cumulant generating function* is defined to be $g(t) = \log M(t)$. Expanding $g(t)$ as a Taylor series

$$g(t) = \sum_{j=1}^{\infty} \frac{c_j}{j!} t^j$$

(the sum starts at $j = 1$ because $g(0) = 0$), the coefficient c_j is called the jth *cumulant* of X. Find the jth cumulant of X, for all $j \geq 1$.

Solution. The MGF of $X \sim \text{Pois}(\lambda)$ is $M(t) = e^{\lambda(e^t - 1)}$, so

$$g(t) = \log M(t) = \log\left(e^{\lambda(e^t-1)}\right) = \lambda(e^t - 1).$$

Using the Taylor series $e^t = \sum_{j=0}^{\infty} \frac{t^j}{j!} = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \cdots$, we have $e^t - 1 = \sum_{j=1}^{\infty} \frac{t^j}{j!}$, so

$$g(t) = \lambda \sum_{j=1}^{\infty} \frac{t^j}{j!} = \sum_{j=1}^{\infty} \frac{\lambda}{j!} t^j = \sum_{j=1}^{\infty} \frac{\lambda}{j!} t^j.$$

Comparing this with $g(t) = \sum_{j=1}^{\infty} \frac{c_j}{j!} t^j$ term by term (matching coefficients of t^j), we need $\frac{c_j}{j!} = \frac{\lambda}{j!}$, so

$$c_j = \lambda \qquad \text{for all } j \geq 1.$$

Q4 — Book Ch.6, Exercise 21

Let $X_n \sim \text{Bin}(n, p_n)$ for all $n \geq 1$, where np_n is a constant $\lambda > 0$ for all n (so $p_n = \lambda/n$). Let $X \sim \text{Pois}(\lambda)$. Show that the MGF of X_n converges to the MGF of X (this gives another way to see that the $\text{Bin}(n, p)$ distribution can be well-approximated by the $\text{Pois}(\lambda)$ when n is large, p is small, and $\lambda = np$ is moderate).

Solution. The MGF of $X_n \sim \text{Bin}(n, p_n)$ is

$$E\left(e^{tX_n}\right) = \left(1 - p_n + p_n e^t\right)^n = \left(1 + p_n(e^t - 1)\right)^n.$$

Substituting $p_n = \lambda/n$:

$$E\left(e^{tX_n}\right) = \left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n.$$

Using the compound-interest limit $\left(1 + \frac{x}{n}\right)^n \to e^x$ as $n \to \infty$, with $x = \lambda(e^t - 1)$ (a constant not depending on n):

$$\left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n \longrightarrow e^{\lambda(e^t-1)}.$$

This is exactly the MGF of $X \sim \text{Pois}(\lambda)$, i.e.

$$E\left(e^{tX_n}\right) \longrightarrow e^{\lambda(e^t-1)} = E\left(e^{tX}\right). \qquad \blacksquare$$

2 Strategic Practice

2. Moment Generating Functions (MGFs)

Q5 — SP 6 §2, Problem 1

Find $E(X^3)$ for $X \sim \text{Expo}(\lambda)$ using the MGF of X (see also Problem 2 in the Exponential Distribution section).

Solution. The MGF of $X \sim \text{Expo}(\lambda)$ is

$$M(t) = E\big(e^{tX}\big) = \left(1 - \frac{t}{\lambda}\right)^{-1} = \frac{\lambda}{\lambda - t}, \qquad t < \lambda.$$

Method 1: Differentiation. Take the third derivative of $M(t)$ and evaluate at $t = 0$.

$$\begin{aligned} M(t) &= \left(1 - \frac{t}{\lambda}\right)^{-1} \\ M'(t) &= \frac{1}{\lambda}\left(1 - \frac{t}{\lambda}\right)^{-2} \\ M''(t) &= \frac{2}{\lambda^2}\left(1 - \frac{t}{\lambda}\right)^{-3} \\ M'''(t) &= \frac{6}{\lambda^3}\left(1 - \frac{t}{\lambda}\right)^{-4}. \end{aligned}$$

Evaluating at $t = 0$:

$$E(X^3) = M'''(0) = \frac{6}{\lambda^3}\,(1-0)^{-4} = \frac{6}{\lambda^3}.$$

Method 2: Pattern recognition via geometric series. Note that $M(t)$ is a geometric series:

$$\frac{1}{1 - \frac{t}{\lambda}} = \sum_{n=0}^{\infty} \left(\frac{t}{\lambda}\right)^n = \sum_{n=0}^{\infty} \frac{t^n}{\lambda^n} = \sum_{n=0}^{\infty} \frac{n!}{\lambda^n} \cdot \frac{t^n}{n!}, \qquad |t| < \lambda.$$

Since the MGF always has the Taylor expansion $M(t) = \sum_{n=0}^{\infty} E(X^n)\frac{t^n}{n!}$, matching coefficients of $\frac{t^n}{n!}$ gives

$$E(X^n) = \frac{n!}{\lambda^n} \qquad \text{for all nonnegative integers } n.$$

In particular, for $n = 3$:

$$E(X^3) = \frac{3!}{\lambda^3} = \frac{6}{\lambda^3}.$$

This second method gives *all* moments of X at once, without needing repeated differentiation.

Q6 — SP 6 §2, Problem 2

If X has MGF $M(t)$, what is the MGF of $-X$? What is the MGF of $a + bX$, where a and b are constants?

Solution. The MGF of $-X$ is obtained by substituting $-X$ for X in the definition:

$$E\left(e^{t(-X)}\right) = E\left(e^{(-t)X}\right) = M(-t).$$

The MGF of $a + bX$ is

$$E\left(e^{t(a+bX)}\right) = E\left(e^{at+btX}\right) = E\left(e^{at} \cdot e^{btX}\right).$$

Since e^{at} is a constant with respect to the expectation over X, it factors out:

$$E\left(e^{at} \cdot e^{btX}\right) = e^{at}\, E\left(e^{btX}\right) = e^{at}M(bt).$$

So the MGF of $-X$ is $M(-t)$, and the MGF of $a + bX$ is $e^{at}M(bt)$.

Q7 — SP 6 §2, Problem 3

Let $U_1, U_2, \ldots, U_{60}$ be i.i.d. Uniform(0,1) and $X = U_1 + U_2 + \cdots + U_{60}$. Find the MGF of X.

Solution. This is the same setup as Q2. The MGF of a single U_1 is, for $t \neq 0$:

$$E\left(e^{tU_1}\right) = \int_0^1 e^{tu}\, du = \left[\frac{e^{tu}}{t}\right]_0^1 = \frac{e^t - 1}{t},$$

and $E(e^{0 \cdot U_1}) = 1$ for $t = 0$.

By independence, the MGF of the sum $X = U_1 + \cdots + U_{60}$ is the product:

$$E\left(e^{tX}\right) = E\left(e^{t(U_1+\cdots+U_{60})}\right) = \left(E\left(e^{tU_1}\right)\right)^{60}.$$

So the MGF of X is 1 at $t = 0$, and for $t \neq 0$:

$$E\left(e^{tX}\right) = \left(\frac{e^t - 1}{t}\right)^{60} = \frac{(e^t - 1)^{60}}{t^{60}}.$$

Q8 — SP 6 §2, Problem 4

Let $X \sim \text{Pois}(\lambda)$, and let $M(t)$ be the MGF of X. The cumulant generating function is defined to be $g(t) = \ln M(t)$. Expanding $g(t)$ as a Taylor series

$$g(t) = \sum_{j=1}^{\infty} \frac{c_j}{j!} t^j$$

(the sum starts at $j = 1$ because $g(0) = 0$), the coefficient c_j is called the jth cumulant of X. Find the jth cumulant of X, for all $j \geq 1$.

Solution. This is the same problem as Q3. The MGF of $X \sim \text{Pois}(\lambda)$ is $M(t) = e^{\lambda(e^t-1)}$, so

$$g(t) = \ln M(t) = \ln\left(e^{\lambda(e^t-1)}\right) = \lambda(e^t - 1).$$

Using the Taylor series $e^t - 1 = \sum_{j=1}^{\infty} \frac{t^j}{j!}$:

$$g(t) = \lambda \sum_{j=1}^{\infty} \frac{t^j}{j!} = \sum_{j=1}^{\infty} \frac{\lambda}{j!} t^j.$$

Matching coefficients with $g(t) = \sum_{j=1}^{\infty} \frac{c_j}{j!} t^j$ gives $\frac{c_j}{j!} = \frac{\lambda}{j!}$, so

$$c_j = \lambda \qquad \text{for all } j \geq 1.$$

3 Homework

Q9 — HW 6, Problem 6

Let $X_n \sim \text{Bin}(n, p_n)$ for all $n \geq 1$, where np_n is a constant $\lambda > 0$ for all n (so $p_n = \lambda/n$). Let $X \sim \text{Pois}(\lambda)$. Show that the MGF of X_n converges to the MGF of X (this gives another way to see that the $\text{Bin}(n, p)$ distribution can be well-approximated by the $\text{Pois}(\lambda)$ when n is large, p is small, and $\lambda = np$ is moderate).

Solution. This is the same problem as Q4. The MGF of $X_n \sim \text{Bin}(n, p_n)$ is

$$E\left(e^{tX_n}\right) = \left(1 - p_n + p_n e^t\right)^n = \left(1 + p_n(e^t - 1)\right)^n.$$

Substituting $p_n = \lambda/n$:

$$E\left(e^{tX_n}\right) = \left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n.$$

Using the fact that $\left(1 + \frac{x}{n}\right)^n \to e^x$ as $n \to \infty$ (which can also be obtained by taking logs and applying L'Hôpital's Rule), with $x = \lambda(e^t - 1)$:

$$\left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n \longrightarrow e^{\lambda(e^t - 1)} = E\left(e^{tX}\right). \qquad \blacksquare$$

Q10 — HW 6, Problem 7

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = e^X$. Then Y has a *Log-Normal* distribution (which means "log is Normal"; note that "log of a Normal" doesn't make sense since Normals can be negative).

Find the mean and variance of Y using the MGF of X, without doing any integrals. Then for $\mu = 0, \sigma = 1$, find the nth moment $E(Y^n)$ (in terms of n).

Solution. The MGF of $X \sim \mathcal{N}(\mu, \sigma^2)$ is $M_X(t) = E(e^{tX}) = e^{\mu t + \sigma^2 t^2/2}$.

Mean of Y: Since $Y = e^X$, we have $E(Y) = E(e^X) = M_X(1)$, i.e. plug in $t = 1$:

$$E(Y) = E\left(e^X\right) = M_X(1) = e^{\mu(1) + \sigma^2(1)^2/2} = e^{\mu + \sigma^2/2}.$$

Second moment of Y: Since $Y^2 = e^{2X}$, we have $E(Y^2) = E(e^{2X}) = M_X(2)$, i.e. plug in $t = 2$:

$$E(Y^2) = E\left(e^{2X}\right) = M_X(2) = e^{\mu(2) + \sigma^2(2)^2/2} = e^{2\mu + 2\sigma^2}.$$

Variance of Y:

$$\begin{aligned}\text{Var}(Y) = E(Y^2) - (E(Y))^2 &= e^{2\mu + 2\sigma^2} - \left(e^{\mu + \sigma^2/2}\right)^2 \\ &= e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2} \\ &= e^{2\mu + \sigma^2}\left(e^{\sigma^2} - 1\right).\end{aligned}$$

(In the last step we factored out $e^{2\mu+\sigma^2}$, since $e^{2\mu+2\sigma^2} = e^{2\mu+\sigma^2} \cdot e^{\sigma^2}$.)

nth moment when $\mu = 0, \sigma = 1$: Here $Y^n = e^{nX}$, so $E(Y^n) = E(e^{nX}) = M_X(n)$, i.e. plug in $t = n$ with $\mu = 0, \sigma = 1$:

$$E(Y^n) = M_X(n) = e^{(0)(n)+(1)^2 n^2/2} = e^{n^2/2}.$$

Q11 — HW 9, Problem 3

Consider independent Bernoulli trials with probability p of success for each. Let X be the number of failures incurred before getting a total of r successes.

(a) Determine what happens to the distribution of $\dfrac{p}{1-p}X$ as $p \to 0$, using MGFs; what is the PDF of the limiting distribution, and its name and parameters if it is one we have studied?

Hint: start by finding the Geom(p) MGF. Then find the MGF of $\dfrac{p}{1-p}X$, and use the fact that if the MGFs of r.v.s Y_n converge to the MGF of a r.v. Y, then the CDFs of the Y_n converge to the CDF of Y.

(b) Explain intuitively why the result of (a) makes sense.

Solution. (a) Setting up. Let $q = 1 - p$. Note that $X \sim \text{NBin}(r, p)$ (negative binomial: number of failures before the rth success).

Step 1: MGF of Geom(p). For $G \sim \text{Geom}(p)$, $P(G = k) = q^k p$ for $k = 0, 1, 2, \ldots$, so

$$E\left(e^{tG}\right) = \sum_{k=0}^{\infty} e^{tk} q^k p = p \sum_{k=0}^{\infty} \left(qe^t\right)^k = \frac{p}{1 - qe^t}, \qquad \text{for } qe^t < 1.$$

This used the geometric series formula $\sum_{k=0}^{\infty} x^k = \dfrac{1}{1-x}$ with $x = qe^t$.

Step 2: MGF of NBin(r, p). Since X (the number of failures before r successes) is a sum of r i.i.d. Geom(p) random variables (the number of failures between consecutive successes), independence gives

$$E\left(e^{tX}\right) = \left(\frac{p}{1 - qe^t}\right)^r, \qquad \text{for } qe^t < 1.$$

Step 3: MGF of $\frac{p}{q}X$. Replacing t with $\dfrac{tp}{q}$ in the MGF of X:

$$E\left(e^{\frac{tp}{q}X}\right) = \frac{p^r}{\left(1 - qe^{tp/q}\right)^r}, \qquad \text{for } qe^{tp/q} < 1.$$

Step 4: Take the limit as $p \to 0$, first for $r = 1$. As $p \to 0$: the numerator $p \to 0$, and the denominator's exponent $qe^{tp/q} \to (1)e^0 = 1$, so the denominator $1 - qe^{tp/q} \to 0$ too — we have a $\frac{0}{0}$ form, so apply L'Hôpital's Rule (differentiating numerator and denominator with respect to p):

$$\lim_{p\to 0} \frac{p}{1 - (1-p)e^{tp/(1-p)}} = \lim_{p\to 0} \frac{1}{e^{tp/(1-p)} - (1-p)\, t \left(\dfrac{1 - p + p}{(1-p)^2}\right) e^{tp/(1-p)}}.$$

As $p \to 0$: $e^{tp/(1-p)} \to e^0 = 1$, and $\dfrac{1-p+p}{(1-p)^2} = \dfrac{1}{(1-p)^2} \to 1$, so the denominator becomes

$$1 - (1)(t)(1)(1) = 1 - t.$$

Therefore

$$\lim_{p \to 0} \frac{p}{1-(1-p)e^{tp/(1-p)}} = \frac{1}{1-t}.$$

Step 5: General r. Since the MGF for general r is the rth power of the $r = 1$ expression raised appropriately (from Step 3, the whole expression is the $r = 1$ result raised to the rth power once we identify $\frac{p}{1-qe^{tp/q}}$ as the $r = 1$ case), for any fixed $r > 0$ as $p \to 0$:

$$E\left(e^{\frac{tp}{q}X}\right) = \frac{p^r}{\left(1-qe^{tp/q}\right)^r} \longrightarrow \left(\frac{1}{1-t}\right)^r = \frac{1}{(1-t)^r}.$$

Step 6: Identify the limit. $\dfrac{1}{(1-t)^r}$ is the MGF of the Gamma$(r, 1)$ distribution, valid for $t < 1$. (One can check the domain condition $qe^{tp/q} < 1$ is equivalent to $t < -\dfrac{1-p}{p}\log(1-p)$, and by L'Hôpital's Rule $\dfrac{-p}{\log(1-p)} \to 1$ as $p \to 0$, so this domain condition converges to $t < 1$, consistent with the Gamma$(r, 1)$ MGF's domain.)

Thus, as $p \to 0$, the scaled Negative Binomial $\dfrac{p}{1-p}X$ converges in distribution to Gamma$(r, 1)$, whose PDF is

$$f(x) = \frac{x^{r-1}e^{-x}}{(r-1)!}, \qquad x > 0.$$

(b) Intuition. The result makes sense because the Gamma distribution is the continuous analogue of the Negative Binomial, just as the Exponential is the continuous analogue of the Geometric. To convert from discrete to continuous, imagine performing many, many Bernoulli trials, each executed very quickly and each having a very low chance of success. To balance the rate of trials against the (small) chance of success per trial, we rescale by $\dfrac{p}{q}$, since this scaling ensures

$$E\left(\frac{p}{q}X\right) = r,$$

which matches the mean of the Gamma$(r, 1)$ distribution. In the limit, the discrete waiting time (measured in number of trials) becomes a continuous waiting time, and the Negative Binomial becomes the Gamma.